

Q1) Guest Master View

- Create a view vw_GuestMaster that returns:
- GuestID, FullName, NationalID/Passport, Phone, Email, City
- TotalStays (count of completed reservations)
- LifetimeSpend (sum of paid amounts) Then query the Top 20 guests by LifetimeSpend.

	Results	Messages							
	GuestID	FullName	NationalID_Passport	Phone	Email	City	TotalStays	LifetimeSpend	
1	9	Isabella Martinez	PASS-1009	+34-91-1234567	isabella.m@gmail.com	Madrid	1	2400.00	
2	36	Christopher Green	PASS-1036	+1-702-555-0366	chris.g@gmail.com	Las Vegas	1	2200.00	
3	18	Sebastian Lee	PASS-1018	+82-2-123-4567	sebastian.l@seoul.kr	Seoul	1	1900.00	
4	4	Olivia Brown	PASS-1004	+61-411-000-111	olivia.b@gmail.com	Sydney	1	1800.00	
5	32	Joseph Wright	PASS-1032	+1-617-555-0322	joseph.w@gmail.com	Boston	1	1800.00	
6	44	Dylan Roberts	PASS-1044	+1-916-555-0444	dylan.r@yahoo.com	Sacramento	1	1800.00	
7	24	Daniel Clark	PASS-1024	+1-904-555-0244	daniel.c@outlook.com	Jacksonville	1	1700.00	
8	39	Lily Gonzalez	PASS-1039	+1-502-555-0399	lily.g@gmail.com	Louisville	1	1700.00	
9	14	Henry Thomas	PASS-1014	+61-3-9876-5432	henry.t@melbourne.com	Melbourne	1	1600.00	
10	33	Chloe Scott	PASS-1033	+1-615-555-0333	chloe.s@outlook.com	Nashville	1	1600.00	
11	47	Nora Campbell	PASS-1047	+1-757-555-0477	nora.c@gmail.com	Virginia Beach	1	1600.00	
12	22	Owen Harris	PASS-1022	+1-408-555-0222	owen.h@gmail.com	San Jose	1	1500.00	
13	45	Addison Turner	PASS-1045	+1-816-555-0455	addison.t@gmail.com	Kansas City	1	1500.00	
14	7	James Miller	PASS-1007	+49-30-123456	james.m@web.de	Berlin	1	1400.00	
15	37	Zoe Adams	PASS-1037	+1-503-555-0377	zoe.a@outlook.com	Portland	1	1400.00	
16	15	Amelia Moore	PASS-1015	+1-602-555-0166	amelia.m@gmail.com	Phoenix	1	1300.00	
17	27	Victoria Robinson	PASS-1027	+1-704-555-0277	victoria.r@gmail.com	Charlotte	1	1300.00	
18	48	Caleb Parker	PASS-1048	+1-404-555-0488	caleb.p@gmail.com	Atlanta	1	1300.00	
19	21	Aria White	PASS-1021	+1-214-555-0211	aria.w@yahoo.com	Dallas	1	1200.00	
20	41	Hannah Carter	PASS-1041	+1-505-555-0411	hannah.c@gmail.com	Albuquerque	1	1200.00	

Q2) Room Availability View (By Date)

- Create a view vw_RoomAvailability that shows for each room:

RoomID, RoomNumber, RoomType, NightlyRate

	RoomID	RoomNumber	RoomType	NightlyRate	CheckInDate	CheckOutDate	ReservationStatus
1	26	501	Penthouse	1200.00	2026-03-01	2026-03-10	Checked-in
2	17	302	Triple	250.00	2026-03-05	2026-03-08	Booked
3	18	303	Triple	250.00	2026-03-12	2026-03-15	Booked
4	4	104	Single	100.00	2026-04-01	2026-04-05	Booked
5	24	404	Suite	500.00	2026-03-15	2026-03-20	Booked
6	20	305	Triple	250.00	2026-03-20	2026-03-25	Booked
7	31	109	Single	105.00	2026-04-10	2026-04-15	Booked
8	33	211	Double	190.00	2026-03-01	2026-03-05	Checked-in
9	36	214	Double	190.00	2026-05-01	2026-05-05	Booked
10	39	307	Triple	260.00	2026-03-10	2026-03-15	Booked
11	43	406	Suite	550.00	2026-04-20	2026-04-25	Booked
12	45	408	Suite	550.00	2026-03-05	2026-03-08	Checked-in
13	48	503	Suite Deluxe	700.00	2026-06-01	2026-06-05	Booked
14	11	206	Double	180.00	2026-03-25	2026-03-30	Booked
15	15	210	Double	180.00	2026-03-01	2026-03-05	Checked-in

- AvailabilityStatus for a given date range (Available / Booked) Then write a query to list all available rooms between two input dates.

Results Messages				
	RoomID	RoomNumber	RoomType	NightlyRate
1	4	104	Single	100.00

Q3) Daily Occupancy Rate

Write a query that calculates occupancy rate per day for a date range:

Date

TotalRooms

OccupiedRooms

OccupancyRate% = OccupiedRooms / TotalRooms * 100

Results Messages				
	Date	TotalRooms	OccupiedRooms	Occupancy Rate %
1	2026-01-01	50	2	4
2	2026-01-02	50	2	4
3	2026-01-03	50	2	4
4	2026-01-04	50	2	4
5	2026-01-05	50	4	8
6	2026-01-06	50	4	8
7	2026-01-07	50	4	8
8	2026-01-08	50	2	4

Q4) Reservation Details View

Create a view vw_ReservationDetails that returns:

ReservationID, GuestName, RoomNumber, RoomType

CheckInDate, CheckOutDate, Nights

ReservationStatus (Booked/Checked-in/Checked-out/Cancelled)

TotalRoomCharge (Nights × NightlyRate)

Results Messages									
	ReservationID	GuestName	RoomNumber	RoomType	CheckInDate	CheckOutDate	Nights	ReservationStatus	TotalRoomCharge
1	1	Liam Smith	101	Single	2026-01-01	2026-01-05	4	Checked-out	400.00
2	2	Emma Johnson	201	Double	2026-01-10	2026-01-15	5	Checked-out	900.00
3	3	Noah Williams	301	Triple	2026-02-01	2026-02-03	2	Cancelled	500.00
4	4	Olivia Brown	401	Suite	2026-01-20	2026-01-25	5	Checked-out	2500.00
5	5	William Jones	501	Penthouse	2026-03-01	2026-03-10	9	Checked-in	10800.00
6	6	Sophia Garcia	102	Single	2026-02-10	2026-02-12	2	Checked-out	200.00
7	7	James Miller	202	Double	2025-12-01	2025-12-05	4	Checked-out	720.00
8	8	Ava Davis	302	Triple	2026-03-05	2026-03-08	3	Booked	750.00
9	9	Isabella Martinez	402	Suite	2026-02-15	2026-02-20	5	Checked-out	2500.00
10	10	Benjamin Rodriguez	502	Penthouse	2026-01-05	2026-01-08	3	Checked-out	3600.00
11	11	Mia Wilson	103	Single	2026-02-22	2026-02-25	3	Cancelled	300.00
12	12	Lucas Anderson	203	Double	2026-01-12	2026-01-14	2	Checked-out	360.00
13	13	Charlotte Taylor	303	Triple	2026-03-12	2026-03-15	3	Booked	750.00
14	14	Henry Thomas	403	Suite	2026-01-18	2026-01-22	4	Checked-out	2000.00
15	15	Amelia Moore	106	Single	2025-11-20	2025-11-25	5	Checked-out	525.00
16	16	Alexander Jackson	104	Single	2026-04-01	2026-04-05	4	Booked	400.00
17	17	Evelyn Martin	204	Double	2026-02-05	2026-02-10	5	Checked-out	900.00
18	18	Sebastian Lee	304	Triple	2026-01-25	2026-01-30	5	Checked-out	1250.00
19	19	Harper Perez	404	Suite	2026-03-15	2026-03-20	5	Booked	2500.00
20	20	Jack Thompson	107	Single	2026-01-10	2026-01-12	2	Checked-out	210.00
21	21	Aria White	105	Single	2026-02-01	2026-02-05	4	Checked-out	400.00
22	22	Owen Harris	205	Double	2025-10-10	2025-10-15	5	Checked-out	900.00
23	23	Ella Sanchez	305	Triple	2026-03-20	2026-03-25	5	Booked	1250.00

Q5) Cancellation Analysis

Write a query showing cancellation performance by month:

YearMonth

TotalReservations

CancelledReservations

Results Messages				
	YearMonth	TotalReservations	CancelledReservations	CancellationRatePercentage
1	2025-10	1	0	0
2	2025-11	2	0	0
3	2025-12	1	0	0
4	2026-01	18	0	0
5	2026-02	14	3	21
6	2026-03	11	1	9
7	2026-04	4	1	25
8	2026-05	2	1	50
9	2026-06	2	1	50

CancellationRate% Also list the Top 5 cancellation reasons (if stored).

	CancellationReason	NumberOfCancellations
1	Personal emergency	1
2	No show	1
3	Illness	1
4	Flight cancelled	1
5	Family reasons	1

Q6) Services Revenue View (Room Service / Laundry / Spa)

Create a view vw_ServiceRevenue that returns:

ServiceName, YearMonth

TotalUsageCount

Results		Messages		
	ServiceName	YearMonth	TotalUsageCount	TotalServiceRevenue
1	Laundry	2025-10	1	200.00
2	Room Service	2025-10	1	150.00
3	Airport Transfer	2025-11	1	250.00
4	Room Service	2025-11	2	300.00
5	Airport Transfer	2025-12	1	250.00
6	Laundry	2025-12	1	200.00
7	Airport Transfer	2026-01	4	1000.00
8	Laundry	2026-01	4	800.00
9	Mini Bar	2026-01	5	1500.00
10	Room Service	2026-01	10	1500.00
11	Spa & Massage	2026-01	5	4000.00
12	Airport Transfer	2026-02	3	750.00
13	Laundry	2026-02	3	600.00
14	Mini Bar	2026-02	4	1200.00
15	Room Service	2026-02	1	150.00
16	Spa & Massage	2026-02	4	3200.00

TotalServiceRevenue Then query Top 5 services by revenue for the last 3 months.

	ServiceName	YearMonth	TotalUsageCount	TotalServiceRevenue
1	Spa & Massage	2026-01	5	4000.00
2	Spa & Massage	2026-02	4	3200.00
3	Room Service	2026-01	10	1500.00
4	Mini Bar	2026-01	5	1500.00
5	Mini Bar	2026-02	4	1200.00

Q7) Invoice Aging & Outstanding Balances View

Create a view vw_InvoiceAging that shows:

InvoiceID, GuestName, InvoiceDate, TotalAmount, PaidAmount, OutstandingAmount

AgingBucket: 0–7, 8–30, 31–60, 60+ days

	InvoiceID	GuestName	InvoiceDate	TotalAmount	PaidAmount	OutstandingAmount	AgingBucket
1	4	William Jones	2026-03-03	1000.00	0.00	1000.00	0-7 days
2	7	Ava Davis	2026-03-03	1200.00	600.00	600.00	0-7 days
3	12	Charlotte Taylor	2026-03-03	500.00	0.00	500.00	0-7 days
4	18	Harper Perez	2026-03-03	300.00	0.00	300.00	0-7 days
5	25	Jackson Lewis	2026-03-03	1500.00	0.00	1500.00	0-7 days
6	33	Samuel Nguyen	2026-03-03	1100.00	0.00	1100.00	0-7 days
7	54	William Jones	2026-03-03	1000.00	0.00	1000.00	0-7 days
8	57	Ava Davis	2026-03-03	1200.00	600.00	600.00	0-7 days
9	62	Charlotte Taylor	2026-03-03	500.00	0.00	500.00	0-7 days
10	68	Harper Perez	2026-03-03	300.00	0.00	300.00	0-7 days
11	75	Jackson Lewis	2026-03-03	1500.00	0.00	1500.00	0-7 days
12	83	Samuel Nguyen	2026-03-03	1100.00	0.00	1100.00	0-7 days

Then query all outstanding invoices in the 60+ bucket.

	InvoiceID	GuestName	InvoiceDate	TotalAmount	PaidAmount	OutstandingAmount	AgingBucket
1	100	Liam Smith	2025-01-01	5000.00	1000.00	4000.00	60+ days

Q8) Payment Method Breakdown

Write a query that returns (monthly):

YearMonth

TotalPaid

PaidByCash, PaidByCard, PaidOnline, PaidByCompany (if applicable)

Failed/Refunded counts (if stored)

	YearMonth	TotalPaid	PaidByCash	PaidByCard	PaidByOnline	PaidByCompany	FailedCount	RefundedCount
1	2026-06	6100.00	0.00	6100.00	0.00	0.00	0	0
2	2026-05	5700.00	0.00	5700.00	0.00	0.00	0	0
3	2026-04	11100.00	5200.00	5900.00	0.00	0.00	0	0
4	2026-03	38000.00	11600.00	26400.00	0.00	0.00	2	0
5	2026-02	36550.00	12400.00	4100.00	20050.00	0.00	2	0
6	2026-01	51100.00	12400.00	11800.00	26900.00	0.00	0	0
7	2025-12	2800.00	0.00	2800.00	0.00	0.00	0	0
8	2025-11	8100.00	0.00	5400.00	2700.00	0.00	1	0
9	2025-10	3700.00	0.00	3700.00	0.00	0.00	0	0

Q9) Housekeeping & Room Turnover Report

Assume Housekeeping has cleaning logs per room per date. Write a query to compute:

CleaningCount per room (last 30 days)

Average time to clean (EndTime - StartTime) if available

	RoomID	RoomNumber	Room Type	CleaningCount_Last30Days	AvgCleanTimeMinutes	LateCleanings	LatePercentage
1	17	302	Triple	4	37	2	50
2	9	204	Double	3	40	0	0
3	11	206	Double	1	45	0	0
4	13	208	Double	1	45	0	0
5	15	210	Double	1	45	0	0
6	19	304	Triple	1	45	0	0
7	20	305	Triple	1	30	0	0
8	23	403	Suite	1	45	0	0
9	25	405	Suite	1	45	0	0
10	26	501	Penthouse	1	30	0	0
11	30	108	Single	1	30	0	0
12	31	109	Single	1	45	0	0
13	33	211	Double	1	45	0	0
14	34	212	Double	1	30	0	0
15	36	214	Double	1	30	0	0
16	38	306	Triple	1	30	0	0
17	40	308	Triple	1	30	0	0
18	41	309	Triple	1	45	0	0
19	43	406	Suite	1	45	0	0
20	44	407	Suite	1	30	0	0
21	46	409	Suite	1	30	0	0
22	48	503	Suite Del...	1	30	0	0
23	50	111	Economy	1	30	0	0

Rooms cleaned late (after a defined cutoff time) (Bonus: create view vw_HousekeepingPerformance.)

	RoomID	RoomNumber	Room Type	TotalCleanings	AvgCleanTimeMinutes	TotalLateCleanings	LatePercentage	FirstCleaning	LastCleaning
1	3	103	Single	1	45	1	100.000000000000	2026-01-06	2026-01-06
2	7	202	Double	1	45	1	100.000000000000	2025-12-06	2025-12-06
3	12	207	Double	1	30	1	100.000000000000	2026-01-15	2026-01-15
4	22	402	Suite	1	30	1	100.000000000000	2025-10-16	2025-10-16
5	27	502	Penthouse	1	45	1	100.000000000000	2026-01-09	2026-01-09
6	32	110	Single	1	30	1	100.000000000000	2026-01-21	2026-01-21
7	37	215	Double	1	45	1	100.000000000000	2025-11-16	2025-11-16
8	42	310	Triple	1	30	1	100.000000000000	2026-01-06	2026-01-06
9	47	410	Suite	1	45	1	100.000000000000	2026-01-26	2026-01-26
10	17	302	Triple	4	37	2	50.000000000000	2026-02-11	2026-03-15
11	18	303	Triple	1	30	0	0.000000000000	2026-01-31	2026-01-31
12	19	304	Triple	1	45	0	0.000000000000	2026-03-21	2026-03-21
13	20	305	Triple	1	30	0	0.000000000000	2026-03-26	2026-03-26
14	21	401	Suite	1	45	0	0.000000000000	2026-01-26	2026-01-26
15	13	208	Double	1	45	0	0.000000000000	2026-02-21	2026-02-21
16	14	209	Double	1	30	0	0.000000000000	2026-01-11	2026-01-11
17	15	210	Double	1	45	0	0.000000000000	2026-03-06	2026-03-06
18	16	301	Triple	1	30	0	0.000000000000	2026-02-04	2026-02-04
19	8	203	Double	1	30	0	0.000000000000	2026-01-15	2026-01-15
20	9	204	Double	3	40	0	0.000000000000	2026-02-11	2026-03-01
21	10	205	Double	1	30	0	0.000000000000	2025-10-16	2025-10-16
22	11	206	Double	1	45	0	0.000000000000	2026-02-26	2026-02-26
23	4	104	Single	1	30	0	0.000000000000	2026-01-06	2026-01-06
24	5	105	Single	1	45	0	0.000000000000	2026-01-06	2026-01-06
25	6	201	Double	1	30	0	0.000000000000	2026-01-16	2026-01-16
26	38	306	Triple	1	30	0	0.000000000000	2026-04-26	2026-04-26

Q10) Staff Performance View (Reception / Billing)

Create a view vw_StaffPerformance for last 30 days:

StaffName, Role

ReservationsHandled (count)

CheckInsProcessed, CheckOutsProcessed

TotalPaymentsProcessed

	StaffID	StaffName	Role	ReservationsHandled	CheckInsProcessed	CheckOutsProcessed	TotalPaymentsProcessed	TotalRevenueProcessed
1	1	James Taylor	Reception	28	25	19	12	41200.00
2	2	Mary Anderson	Reception	27	25	16	10	36800.00
3	3	Robert Wilson	Billing	0	0	0	0	0.00
4	4	Patricia Garcia	Billing	0	0	0	0	0.00
5	5	Linda Thomas	Housekeeping	0	0	0	0	0.00
6	6	William Moore	Housekeeping	0	0	0	0	0.00
7	7	Barbara Martin	Housekeeping	0	0	0	0	0.00
8	8	Richard Jackson	Housekeeping	0	0	0	0	0.00
9	9	Susan Lee	Housekeeping	0	0	0	0	0.00
10	10	Joseph White	Manager	0	0	0	0	0.00
11	11	Ahmed Hassan	Reception	0	0	0	0	0.00
12	12	Fatma Ali	Housekeeping	0	0	0	0	0.00
13	13	Mohamed Khal...	Billing	0	0	0	0	0.00
14	14	Sara Mostafa	Manager	0	0	0	0	0.00
15	15	Youssef Ibrahim	Reception	0	0	0	0	0.00

TotalRevenueProcessed Then query the Top 10 staff by TotalRevenueProcessed

	StaffName	Role	ReservationsHandled	CheckInsProcessed	CheckOutsProcessed	TotalPaymentsProcessed	TotalRevenueProcessed
1	James Taylor	Reception	28	25	19	12	41200.00
2	Mary Anderson	Reception	27	25	16	10	36800.00
3	Ahmed Hassan	Reception	0	0	0	0	0.00
4	Joseph White	Manager	0	0	0	0	0.00
5	Susan Lee	Housekeeping	0	0	0	0	0.00
6	Richard Jackson	Housekeeping	0	0	0	0	0.00
7	Barbara Martin	Housekeeping	0	0	0	0	0.00
8	William Moore	Housekeeping	0	0	0	0	0.00
9	Linda Thomas	Housekeeping	0	0	0	0	0.00
10	Patricia Garcia	Billing	0	0	0	0	0.00